Supplementary File S2: Codebook

Section references and author-year citations refer to the main manuscript. The codebook covers RQ1–RQ4 and includes study-linked coding examples.

This codebook documents the classification scheme applied to the 78 included studies. Each dimension is defined, its categories are given decision rules, and worked examples drawn from the coded corpus are provided. A study could receive more than one code on a dimension where it exhibited more than one qualifying attribute; the resulting denominators are 94 coded occurrences for RQ1, 123 for RQ2, 83 for RQ3 and 93 for RQ4.

Coding was performed independently by two reviewers using these rules. Disagreements were resolved by discussion, with a third author consulted where consensus could not be reached. Agreement was not quantified with a formal reliability statistic.

General rules

1. A code is assigned only where the study text supports it explicitly. Attributes inferred from the name of a platform alone are not coded.
2. Where a study describes several platforms or interventions, each is coded separately, producing multiple coded occurrences for that study.
3. Where a study reports several outcomes or instruments, each outcome–instrument pair is coded separately.
4. Residual categories are used only after all substantive categories have been excluded, and never as a default.

RQ1 dimensions

Category	Decision rule	Example from the corpus
Platform type (one code per intervention)		
LMS	Centralised system hosting structured courses with enrolment and activity tracking.	“LMS with modular design; basic interactivity”
Mobile App	Delivery through a dedicated application on a personal mobile device.	“Mobile app with self-paced modules”
Web Portal	Browser-delivered content without LMS enrolment or tracking structure.	Web-delivered resource repository
VR/Virtual Patient	Immersive environment or simulated patient encounter with interactive clinical decisions.	“VR for procedural simulation”

Category	Decision rule	Example from the corpus
Gamified Platform	System whose primary organising principle is game mechanics (points, levels, leaderboards).	"Gamified platform with scoreboards"
Teaching method		
Self-paced learning	Learner controls sequence and timing; no scheduled cohort progression.	"Mobile app with self-paced modules"
Case-based learning	Instruction organised around analysis of clinical cases without role assignment.	"Online modules with case-based design"
Role-based training	Content or pathway differentiated by the learner's professional role.	"LMS with quizzes and structured content" delivered by role
Simulation-based learning	Learning through controlled reproduction of a clinical situation requiring performed action.	"VR for procedural simulation"
Pedagogical features (multiple codes permitted)		
Gamification	Points, badges, missions or leaderboards used as motivational devices.	Scoreboard-driven progression
Immediate feedback	Automated response delivered on completion of an exercise or test item.	Post-quiz automated correction
Interactivity	Learner choices materially influence the learning sequence or its outcome.	Branching clinical decision path
Spaced repetition	Content deliberately re-presented at timed intervals.	Scheduled review of prior items
Technological features (multiple codes permitted)		
Modularity	Content divided into independently completable units.	"LMS with modular design"
Multi-device/mobile accessibility	Explicitly usable across device classes including mobile.	"Mobile app, flexible access"
Progress tracking	System records and displays learner advancement.	Completion dashboard
Adaptive learning	Content or difficulty adjusts automatically to learner performance.	Performance-driven path adjustment
Multilingual accessibility	Content available in more than one language.	Localised content versions

RQ2 dimensions

Outcome domains follow the Kirkpatrick taxonomy (Kirkpatrick, 1994) as adapted for healthcare education. For the Outcome ↔ Tool analysis (section 4.3) the detailed outcome codes are grouped as follows: *Cognitive* comprises Learning; *Behavioural* comprises Performance, Behaviour, Skill acquisition and Clinical practice; *Affective*

comprises Engagement, Satisfaction, Motivation, Perception and Confidence; *Organisational/Clinical* comprises Clinical outcomes, Efficiency and Time. Instrument codes are grouped as: *Objective tests*; *Platform analytics* (platform data); *Performance-based* (OSCE, simulations, checklists, rating scales, expert and external evaluation); and *Self-report surveys*.

Category	**Decision rule**	**Example from the corpus**
Learning	Change in knowledge demonstrated by assessment.	“Improved theoretical knowledge”
Performance	Measured execution of a task under assessment conditions.	Procedural score on a rated task
Behaviour	Observed change in practice beyond the assessment setting.	Protocol adherence in the workplace
Skill acquisition	Attainment of a specific technical competence.	Suturing proficiency attained
Clinical practice	Reported change in routine clinical activity.	Change in documented practice
Engagement	Participation intensity or sustained use.	Course completion and activity levels
Satisfaction	Learner-reported contentment with the intervention.	“User satisfaction”
Motivation	Learner-reported drive to engage.	Reported motivation change
Perception	Learner-reported views or attitudes about the intervention.	“Increased learning flexibility”
Confidence	Learner-reported self-efficacy.	Self-rated preparedness
Clinical outcomes	Patient-level results.	Patient-level effect reported
Efficiency	Resource or process gain.	Reduced training resource use
Time	Duration measures.	Time to complete a procedure
Objective Tests	Scored knowledge instruments including pre- and post-tests.	“Pre- and post-tests”
Platform Data	Interaction traces captured by the system.	Logged activity metrics
OSCE	Structured clinical examination.	OSCE station scores
Simulations	Assessment performed within a simulated encounter.	Simulated scenario rating
Checklists	Itemised observational instrument.	Procedural checklist
Rating scales	Graded observational or self-report scale.	Likert-type competence scale
Surveys	Questionnaire capturing self-reported constructs.	“Likert-scale surveys”

Category	Decision rule	Example from the corpus
Expert / External Evaluation	Judgement by an expert or an assessor external to the programme.	External assessor judgement

RQ3 dimensions

Context Tag records the setting in which the intervention was delivered: Healthcare environment (clinical service settings), Academic environment (university teaching outside a clinical service), Nursing School, Surgical Training programme, CME (formal continuing education provision), Simulation Center (dedicated simulation facility), Allied Health, and Remote/Telehealth. Where a study spans settings, the setting in which the intervention was delivered takes precedence over the affiliation of the authors.

Role Tag records the professional status of participants: Undergraduate Students (Medical/Nursing); Postgraduate Trainees (residents, specialising physicians); Practicing Professionals (doctors, nurses in service); and Mixed or Interprofessional Groups, assigned where participants span more than one of the preceding categories or where the study reports a combined cohort without disaggregation.

Study-linked illustrations of the decision rules

The following examples illustrate how to apply the codebook to the 78 reports of this study. They make the reasoning behind individual codes explicit; they are not a reconstruction of the original independent coding records or an additional reliability assessment. Information in the second column is paraphrased from the reports. Codes shown across different dimensions describe complementary attributes and do not imply additional occurrences within any one dimension.

Table S2.1. Study-linked illustrations of codebook application.

Report	Reported information	Dimension and code	Decision rationale
Lohre et al. (2020)	The intervention uses immersive virtual reality to rehearse reverse shoulder arthroplasty.	Platform Type: VR/Virtual Patient; Teaching method: Simulation-based learning.	The delivery technology is immersive VR; the instructional task reproduces a surgical procedure. These are separate dimensions.
Lohre et al. (2020)	Participants were recruited during an annual shoulder and elbow resident and fellow training course.	Context Tag: Surgical Training programme.	The code follows the intervention's training setting, rather than the authors' institutional affiliations.
Lohre et al. (2020)	Participants were senior orthopaedic surgery residents in postgraduate years four and five.	Role Tag: Postgraduate Trainees.	Participants were undergoing specialist training; the surgical subject alone does not determine their professional-status code.

Report	Reported information	Dimension and code	Decision rationale
da Silva Negreiros et al. (2022)	E-MunDiabetes is a mobile application developed for diabetes education.	Platform Type: Mobile App.	Delivery through a dedicated mobile application supports this code. No LMS, progress-tracking or adaptive-learning code follows from the application name alone.

RQ4 dimensions

Systemic barriers are organisational, infrastructural or resource-level obstacles external to the individual learner. Individual barriers are learner-level attributes or circumstances. Both dimensions carry the residual category *Lack of Clarity in Barrier Description*.

Category	**Decision rule**	**Example from the corpus**
Systemic barriers		
Infrastructure Limitations	Hardware, network or equipment availability constrains delivery.	“Access to equipment”; “Internet instability”
Funding Constraints	Cost of acquisition or operation is identified as limiting.	“High cost of VR setup”; “Data usage costs”
Poor UX	Overall interaction experience impedes use.	“Difficult content navigation”
UI Design	Interface presentation specifically identified as deficient.	“Outdated interface”
Cultural	Organisational or professional culture resists adoption.	Resistance to changed training practice
Linguistic Barriers	Language of delivery limits access.	“Language localization issues”
Platform Interactivity Issues	Platform affords insufficient interaction.	“Lack of interaction, tech barriers”
Lack of Realistic Scenarios	Scenario fidelity or variety identified as insufficient.	“Limited realism of scenarios”
Lack of Technical Support	Absence of support provision identified.	“Technical support needed”
Lack of Gamified Elements	Absence of motivational mechanics identified as limiting.	“Gamification not suited for all learners”
Lack of Clarity in Barrier Description	A limitation is reported but is generic, or concerns content and instructional design rather than organisational or resource constraints, and cannot be attributed to any category above.	“Poor customization for specialty topics”; “Limited variety of clinical cases”; “Motion sickness in some users”
Individual barriers		

Category	Decision rule	Example from the corpus
Cognitive Overload	Volume, density or pacing of content exceeds learner capacity.	“Content overload”
Time Constraints	Competing demands prevent participation.	“Scheduling conflicts”
Low Digital Literacy	Learner technical skill limits use.	“Digital literacy gaps”
Low Engagement	Observed or reported disengagement during use.	“Low engagement”
Motivation	Learner-reported lack of drive to participate.	“Low learner motivation”
Lack of Clarity in Barrier Description	An obstacle is reported but is not expressed as a learner-level attribute or circumstance, and cannot be attributed to any category above.	“No feedback loop”; “Outdated material”; “Inadequate assessment”

The residual category is not equivalent to “no barrier reported”. In most instances the study reports a difficulty, but one that is either generic or that pertains to a dimension other than the one being coded. Because the category is frequent, section 4.5 reports barrier distributions both over all coded occurrences and restricted to attributable barriers.